

MOBRIDGE, SD, USA

WMO: 726685

Lat: 45.546N

Lon: 100.408W

Elev: 523

StdP: 95.20

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94052

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.5	-22.6	-28.8	0.3	-25.0	-26.2	0.4	-22.3	14.7	-7.1	12.7	-7.1	2.9	310	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	35.0	21.2	32.8	20.9	31.0	20.4	23.9	30.4	22.8	29.4	21.7	28.5	5.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.9	27.5	20.9	16.5	26.4	19.5	15.2	25.3	74.3	30.1	69.9	29.5	66.1	28.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.4	10.2	DB	-30.5	39.0	3.3	3.0	-32.8	41.1	-34.7	42.8	-36.5	44.5	-38.9	46.6	(4)
(5)			WB	-30.6	25.8	3.2	1.2	-32.9	26.7	-34.8	27.4	-36.6	28.1	-38.9	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-8.8	-6.8	-0.2	7.2	13.5	19.3	23.3	22.1	17.0	8.6	0.6	-6.7	(6)	
	DBStd	12.65	7.76	7.62	7.36	5.80	4.77	4.02	3.49	3.51	4.99	5.27	6.23	6.96	(7)	
	HDD10.0	2411	584	471	323	117	21	0	0	0	5	88	286	517	(8)	
	HDD18.3	4358	842	704	576	334	162	35	3	8	83	303	533	776	(9)	
	CDD10.0	1501	0	0	6	33	129	281	413	376	214	46	3	0	(10)	
	CDD18.3	406	0	0	0	1	12	65	158	126	43	2	0	0	(11)	
	CDH23.3	4500	0	0	3	34	182	700	1710	1327	502	42	0	0	(12)	
	CDH26.7	1888	0	0	0	7	52	255	786	583	194	11	0	0	(13)	
(14) Wind	WSAvg	4.7	4.3	4.4	4.9	5.3	5.1	4.7	4.4	4.5	4.8	4.9	4.5	4.4	(14)	
(15) Precipitation	PrecAvg	431	10	11	16	33	70	86	59	54	35	36	11	10	(15)	
	PrecMax	580	31	40	64	115	208	205	137	147	111	136	32	38	(16)	
	PrecMin	206	3	1	3	4	7	20	18	13	0	1	0	0	(17)	
	PrecStd	91	6	8	13	24	44	37	31	32	29	34	11	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	12.7	22.4	27.4	31.1	35.0	39.2	37.5	35.1	27.9	19.1	10.8	(19)	
		MCWB	5.0	6.2	11.4	14.3	18.7	21.0	21.8	21.4	20.2	15.3	10.4	5.3	(20)	
	2%	DB	6.1	8.0	17.4	23.0	27.8	32.0	35.7	34.1	31.5	23.2	15.1	7.4	(21)	
		MCWB	2.7	4.0	9.7	12.3	17.4	20.5	21.9	21.3	19.5	13.8	8.4	3.4	(22)	
	5%	DB	3.6	5.5	13.5	20.0	25.0	29.5	33.0	32.2	28.5	20.0	12.1	4.4	(23)	
		MCWB	1.2	2.5	7.4	11.0	15.8	19.6	21.2	20.7	18.5	12.1	6.5	1.3	(24)	
	10%	DB	1.9	3.0	10.2	17.3	22.5	27.4	31.1	30.0	26.1	17.2	9.0	2.4	(25)	
		MCWB	0.1	0.8	5.8	9.4	14.6	18.8	21.0	19.9	17.3	11.0	5.1	0.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	7.1	12.6	15.4	20.6	24.5	25.7	24.6	22.7	16.8	11.2	5.7	(27)	
		MCDB	9.8	11.1	19.5	24.4	28.0	30.8	32.2	31.6	29.3	24.9	17.5	9.9	(28)	
	2%	WB	2.8	4.3	10.2	13.0	18.8	22.6	24.2	23.0	21.0	14.9	8.8	3.5	(29)	
		MCDB	5.6	7.9	16.8	22.1	25.8	29.1	30.5	30.2	28.8	21.5	14.8	7.1	(30)	
	5%	WB	1.4	2.6	7.9	11.4	16.9	21.1	23.2	21.9	19.7	12.9	6.8	1.7	(31)	
		MCDB	3.5	5.3	13.4	18.7	23.0	27.5	29.7	28.7	26.4	19.2	11.4	4.2	(32)	
	10%	WB	0.0	0.9	5.6	10.0	15.4	20.0	22.2	21.0	18.3	11.4	5.2	0.2	(33)	
		MCDB	1.8	2.9	10.2	16.4	20.9	26.0	28.8	27.9	24.5	16.4	8.9	2.2	(34)	
(35) Mean Daily Temperature Range	MDBR		10.3	10.6	11.4	13.0	12.7	12.4	13.2	13.6	13.5	11.9	10.8	9.7	(35)	
	5% DB	MCDBR	12.5	13.1	17.4	18.4	16.8	15.4	16.4	16.5	16.9	17.2	15.8	13.1	(36)	
		MCWBR	9.6	9.4	10.4	9.7	8.4	7.0	6.1	6.3	7.1	9.0	9.6	9.5	(37)	
	5% WB	MCDBR	12.0	12.5	16.9	16.8	14.7	13.3	13.3	14.2	14.6	15.2	15.1	12.7	(38)	
		MCWBR	9.3	9.1	10.4	9.2	8.2	7.1	6.4	6.5	7.1	8.4	9.5	9.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.258	0.265	0.290	0.326	0.353	0.368	0.376	0.377	0.343	0.307	0.273	0.256	(40)	
	taud		2.487	2.500	2.508	2.414	2.413	2.405	2.413	2.386	2.463	2.533	2.524	2.499	(41)	
	Ebn at Noon		875	936	951	935	914	898	885	871	877	870	848	840	(42)	
	Edh at Noon		65	78	90	109	113	115	112	111	94	76	63	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.74	2.79	3.81	4.95	5.87	6.65	6.81	5.70	4.46	2.85	1.88	1.38	(44)	
	RadStd		0.14	0.27	0.33	0.35	0.54	0.45	0.30	0.47	0.30	0.35	0.22	0.08	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page